

Laser Heating Beamtime Pre-Experiment Questionnaire

General

1. Beamtime Dates

2. Radiological Beamtime?

☐ No ☐ Yes ☐ Partially

3. What technique(s) are we trying to use?

- ☐ Standard DAC (no LH) ☐ Laser-heated DAC
- ☐ Membrane-driven compression
- ☐ Dynamic DAC (dDAC / time-resolved)
- ☐ Resistively heated ☐ Single-crystal / Multi-grain DAC
- ☐ Toroidal DAC

Other (please specify):

4. What DAC types will you be bringing?

- ☐ Standard symmetric DAC (Princeton type)
- ☐ BX-90 ☐ Membrane-driven DAC
- ☐ Dynamic DAC (dDAC) ☐ DAC in vacuum jacket
- ☐ Custom / in-house DAC ☐ Need a custom holder for DAC?

Other (please specify):

Beam Specs

5. Sample chamber sizes (μm) (please list)

6. Required X-ray focus, FWHM ($\mu\text{m} \times \mu\text{m}$)

Typical focus ($\mu\text{m} \times \mu\text{m}$): 1.5x1.5

7. Preferred X-ray energy (keV)

Default energy (keV): 29.2. 16-ID-B uses 29.2 keV for 99% of its experiments, note that requesting a different energy will add some setup time for your beamline scientist.

8. Do you anticipate needing to defocus or change beam size during experiment?

☐ No ☐ Yes

If yes, please describe:

Mirrors

9. Which mirror will be used? (default: LKB)

☐ SKB ☐ LKB

10. Isolation mode? (Only for radiological beamtime)

☐ No ☐ Yes

Expected energy, mirror, or isolation mode changes during beamtime:

Beamtime Daily Plan

Day 1:

dk;lifa;kfasd;jfjkdsa;lfjdas;fkdsa;fkda;fdksf;skldf;dsjfkdsfkdsafjdsfkldsfjdskf;
dsklfjads;jjdsfklf;dsjfkldsf;ksafds;kfjdsa;klfjsad;lkfjas;lkfasj;flkdsjf;lskjdn fsfkds;
dsakllj fl akl;kd f. kldak;fdj;akjf dsf asdk;sdsckfas;ljsdkf s kklfjfldf;lkfalkdj f. dfk;ajksfj
sd;lal fdjaklf ads;flasdk f;lksdaf dsfjs ad;fkl. f;lakldfsaka;ljd; lakdjf;aslf as;jf a ;
ldfja dfjkla fd; a;lkfajd f df sfdsjffjd f dffs f. d;klfjas ;kdlfj a;ld kfj a;ldkf a df; a; d f ff.

Day 2:

Day 3:

Day 4:

Day 5+:

Capabilities

11. Online ruby system?

☐ No

☐ Yes

12. Other high temperature method(s)

☐ Resistive heating

Please describe:

13. Do you need any of the following?

☐ Vacuum pump

☐ Chiller

☐ Thermocouple interface

Other (please specify):

14. How do you want to heat?

☐ Square modulated

☐ Continuous

Please describe: